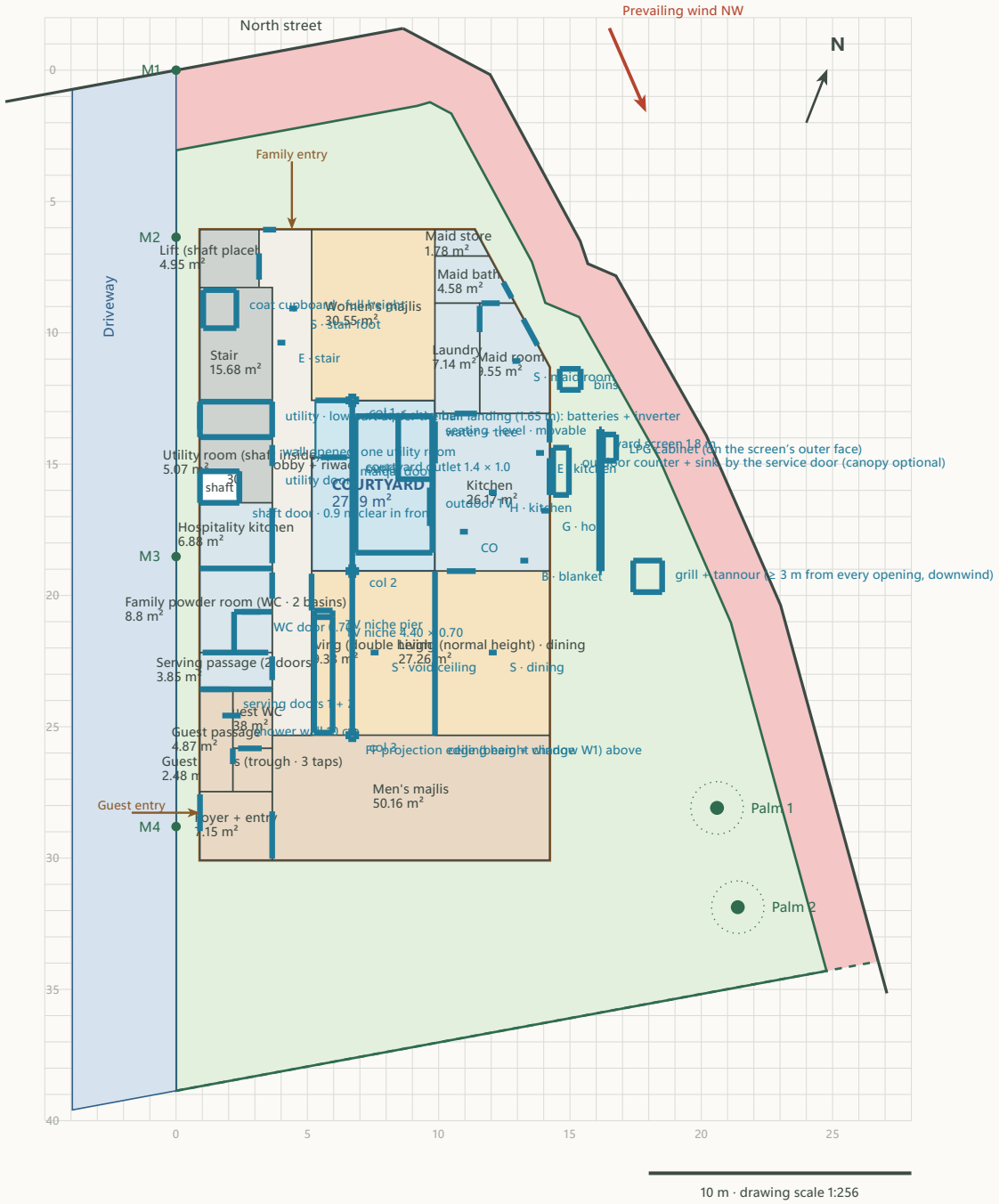


Taif / Palm House

Ground floor V10.0 · pentagon on site option Z4 · the owner's review of 10 Nov 2026

Taif / Palm House V10.0 · base.typ + design_gf.typ + data/rooms_gf.csv · room studies in studies/ · full description docs/PROJECT_V10.md

Ground floor V10.0 on site option Z4, with **three columns** on x 5.8 (in the women's majlis south wall, between the courtyard glass and the living, in the living's south wall) under the edge beam of the first floor's 1.55 m west projection (enclosure option A, the gallery glazed on the edge beam), a colonnade along the courtyard edge with a three-arch arcade: a pentagon 13.311 × 24.0 m on wall centre lines (13.561 × 24.25 outside), the north-east corner cut parallel to the lane down to y 5.245, then the east wall straight south. Outer west wall face 0.79 m from the M line; nearest outer wall face to the envelope 0.91 m (to be confirmed by the official survey). Both palms lie beyond the south facade. Wall allowances 250 mm outer, 200 mm inner, 100 mm dry block partitions. **Middle band: women's majlis** 4.70 × 6.5 · **courtyard** an L-shape round the malqaf chamber, open to the sky, the floor level with movable seating · **living (double height)** 4.70 × 6.25, entered from the riwaq through a 1.40 m arched opening; its west wall solid and full height with the arched TV niche 4.40 × 0.70 m. **The malqaf (V10.0, L5):** its chamber at the courtyard's north-west corner inside the covered strip (the dust trap, the filters and the fan, the space for a cooler; the carved door onto the strip), its **courtyard outlet** 1.4 × 1.0 in the first arcade arch facing the seating, the shaft and the head above. **East service block** (studies/east_wing.typ): **kitchen** 4.361 × 6.0, two parallel counter runs, no island, sink on the courtyard wall, hob and hood on the outside east wall, hood extract straight out through the east wall (no east duct); **laundry** 1.70 × 4.2; **maid room** ≈ 8.4 m² clear and **maid bath** 1.8 m deep up to the cut wall, the **maid's store** in the cut tip; maid access only kitchen → laundry → maid room → maid bath (the maid's wing re-planned in studies/maid_wing.typ, a study). **West band (the owner's review, 10 Nov 2026):** the lift (installed at construction), the open stair (flipped, V8.2; the coat cupboard in its start strip), **one utility room with one door from the riwaq**: the west service **shaft** (x 0–1.5, y 9.2–10.4, drawn black, its maintenance door with 0.9 m clear in front), the panel, the heater and the manifold on the walls, and the **low part under the stair's half landing** (1.65 m) for the future batteries and inverter: the wall between them opened (the engineer), the store's door onto the riwaq removed. **West wash block** (2.75 × 13.6, studies/guest_wing.typ): **hospitality kitchen** 2.75 × 2.5 (open to the riwaq), a full-height wall, the **family powder room** 2.75 × 3.2: the WC in its own compartment (≈ 1.3 × 1.4 clear) with its own door and a small hand basin, and in the open zone in front of the entrance a vanity with **two basins under a large mirror**; no shower. The **serving passage** with its two doors (never open together). **Guest side:** the **guest WC** 1.5 × 2.25 (0.5 m deeper, taken from the basins room) with a **simple shower cubicle** at its north end (a 100 mm block wall, open entrance, wall-mounted shower and mixer, a floor drain with falls, tanked and tiled) and the WC with its hand spray; the **guest basins** 1.5 × 1.65 keep the trough with three taps; the guest passage and the foyer with the double door and the 1.80 m opening into the **men's majlis** 10.561 × 4.75.

✓ **All checks passed**

Areas

Envelope	**667.87 m²**
Building limit (envelope unless set)	**311.99 m²**
Other open voids	**1.8 m²**
Courtyard (open to sky)	**27.19 m²**
Rooms placed (20)	**279.96 m²**
Limit minus courtyard and voids	**283 m²**
Air channel + duct	**3.03 m²**
Not yet assigned	**3.04 m²**

Orientation

True compass vs the M line	**21.4°**
NW wind vs the M line	**23.6°**

Concept only. Wind: Global Wind Atlas (area data, 10 m), NW ≈ 15–20 %, SE ≈ 7–8 %. Confirm on site with a ribbon test.

Schedules

All coordinates in metres from M1: x east (square to the M line), y along the M line southwards.

Envelope corners

Corner	x	y
V1	0	38.86
V2	0	3.05
V3	9.18	1.36
V4	9.67	1.22
V5	10.48	1.65
V6	13.55	7.3
V7	14.06	8.86
V8	15.35	9.4
V9	18.4	14.8
V10	21.13	21.04
V11	24.77	34.3

Rooms

Name	Frame	w × d (m)	Area
Women's majlis	M	4.7 × 6.5	30.55 m²
Living (double height)	M	4.7 × 6.25	29.38 m²
Men's majlis	M	10.561 × 4.75	50.16 m²
Maid bath	M	3.032 × 1.8	4.58 m²
Maid room	M	2.661 × 4.2	9.55 m²
Laundry	M	1.7 × 4.2	7.14 m²
Kitchen	M	4.361 × 6.0	26.17 m²
Living (normal height) · dining	M	4.361 × 6.25	27.26 m²
Lift (shaft placeholder)	M	2.25 × 2.2	4.95 m²
Family lobby + riwaq	M	2.0 × 19.25	30.3 m²
Stair	M	2.75 × 5.7	15.68 m²
Utility room (shaft inside)	M	2.75 × 2.5	5.07 m²
Hospitality kitchen	M	2.75 × 2.5	6.88 m²
Family powder room (WC · 2 basins)	M	2.75 × 3.2	8.8 m²
Serving passage (2 doors)	M	2.75 × 1.4	3.85 m²
Guest WC	M	1.5 × 2.25	3.38 m²
Guest basins (trough · 3 taps)	M	1.5 × 1.65	2.48 m²
Guest passage	M	1.25 × 3.9	4.87 m²
Foyer + entry	M	2.75 × 2.6	7.15 m²
Maid store	M	2.054 × 1.0	1.78 m²

Design idea and philosophy

Taif / Palm House · Ground floor V10.0 · pentagon with the north-east corner cut

A house around a cool heart

The house is organised around one idea: a small, shaded courtyard at its centre that stays cool through the day. The rooms the family uses most open onto it. The wide double-height living room, which also holds the dining, faces it from the south; the kitchen from the east; the women's majlis from the north. On the west, the riwaq runs along it with windows. The courtyard is not a leftover space between rooms; it is the room the rest of the house looks into.

Looking in, not out

The outer walls stay quiet, with small openings, as in the traditional houses of the region. Light, air and view come from the inside. This gives privacy from the street and the lane, protects the rooms from the hot sun and dust, and lets the family live openly within their own walls.

Air before sun

Taif's prevailing wind comes from the north-west. The malqaf (V10.0, L5) stands at the courtyard's north-west corner: an RC core inside the covered strip with its foot chamber (the dust trap, the filters and the fan, the space for a cooler; a carved door onto the strip), the breeze outlet in the first arcade arch facing the seating, the shaft in the gallery's end and the head 1.6 × 2.4 on the roof; the north-east tip is the maid's store, so it takes no usable room space, catches it above the roof, where it is strong and clean, and brings it down to a tunnel below the floor. The tunnel passes through the cool ground and releases the air through a low, closable opening in the courtyard wall. Warm air rises and leaves through the open sky. The narrow courtyard is enclosed on all four sides and shaded by its walls, so it receives air without receiving sun. Every opening can be closed: open at night and early morning, closed in hot afternoons, dust and cold.

On firm ground, as far north as possible

The house sits on the firm, higher ground in the north of the plot, square to the M line so every wall can be set out from the site pegs, and pushed as far north as the setbacks allow. The narrow courtyard keeps the house narrow, and the north-east corner is cut parallel to the lane, so the whole house reaches further north and both palms stay well clear behind it.

One living room, two heights

The living room is one open space with two ceiling heights. The half next to the courtyard is double height: it continues the courtyard's vertical space, gives the best view into it, and works as a chimney, with high operable windows that let warm air rise and leave while cool air is drawn in from the courtyard; its 4.70 m axis width (4.50 m clear) is a simple roof span. The east half is normal height: a warmer, lower place for dining beside the kitchen door, easier to air-condition and light, with ducts hidden in its ceiling, and the first floor continues above it. No upper floor is built over the double-height half or the courtyard.

One band that serves the house

Everything that serves the house sits in one band on the west: the family entrance with the coat cupboard beside the stair's foot, the lift shaft (the lift is installed at construction), the stair, the utility room with the service shaft and the west wash block, one strip of wet rooms on one drain run (hospitality kitchen, family basins, family WC, guest WC, guest basins). Pipes, cables and air rise in one place, the structure gains one stiff core, and every service can be reached for maintenance without breaking a wall. The family lobby and riwaq run along this band as one straight strip on the courtyard side, the family's covered route from the front door to the living room.

Family, guests and service, each in their place

The family enters through the lobby; women guests use the same door to reach the women's majlis. The men's majlis has its own foyer, entered by a double door at one end and opening into the majlis at the other (a bent entry), with a shoe cabinet and hooks for the bisht and shemagh, a trough basin with three taps and a WC whose door hides behind a short screen wall, all in the south of the west wash block; the guest side has no door onto the riwaq. Its only link to the house is a short serving passage with only two doors that are never open together: the family opens door 1 from the riwaq, leaves the tray and closes it, then calls the host, who opens door 2 from the guest passage and collects. Door 1 locks from the riwaq side and door 2 from inside the passage, so the family controls both. An open door 2 shows only the passage and a blank wall, a bent entry (majaz). The hospitality kitchen just north of it, a light kitchen for coffee, tea, dates and desserts, is the serving base; the east kitchen is the heavy one. The living room opens to the riwaq without a door and only a 1.20 m low partition faces the serving door, so the family's privacy rests on the two serving doors never being open together. The kitchen stays isolated beside the courtyard, a clean 4.36 × 6.5 m room on the same lines as the courtyard, with two parallel counter runs, its only door to the dining and an open entrance to the laundry. The laundry runs north along the solid wall of the women's majlis to the maid room and maid bath, which use the sloped corner to the full; the maid wing is reached only through the kitchen. Kitchen, laundry and maid bath drain straight out to the east, to a manhole in the east service yard; the vent stack runs up the outside face of the east wall, and the kitchen extract leaves straight through the east wall at the hood (no east duct).

Services built for life

The service shaft sits inside a utility room, so every service is in one place behind one door: the shaft, the main electrical panel, the water manifold with its isolation valves and the water heater, and (10 Nov 2026, owner) the low part under the stair's half landing as the same room, where the future batteries and inverter go. The shaft is a closed box of about 1.0 × 1.2 m clear with sealed walls and one full-height maintenance door, and the 0.9 m in front of that door is always kept clear. It stands directly beside the west wash block, so drainage runs stay short; first-floor bathrooms should stack over the north part of the block, near the shaft. The block's drain crosses under the serving passage floor, with a cleanout access point there. The space under the stair's half landing is the low part of the utility room (1.65 m headroom), not a separate store. The lift shaft is built now as a store and fitted with a two-stop lift later. The Saudi Building Code sets no fixed size for a closed pipe shaft; the sizes here follow general professional guidance and must be confirmed with the code, the municipality and the plumbing engineer.

Concept for discussion (V10.0). The malqaf's breeze and its filtered supply, the waterproofing, the drainage, the structure and the services need the engineers' review; the wind faces to be confirmed by the on-site weather station.